BIOE4072.1 PLANNING AND MANAGEMENT OF RESEARCH

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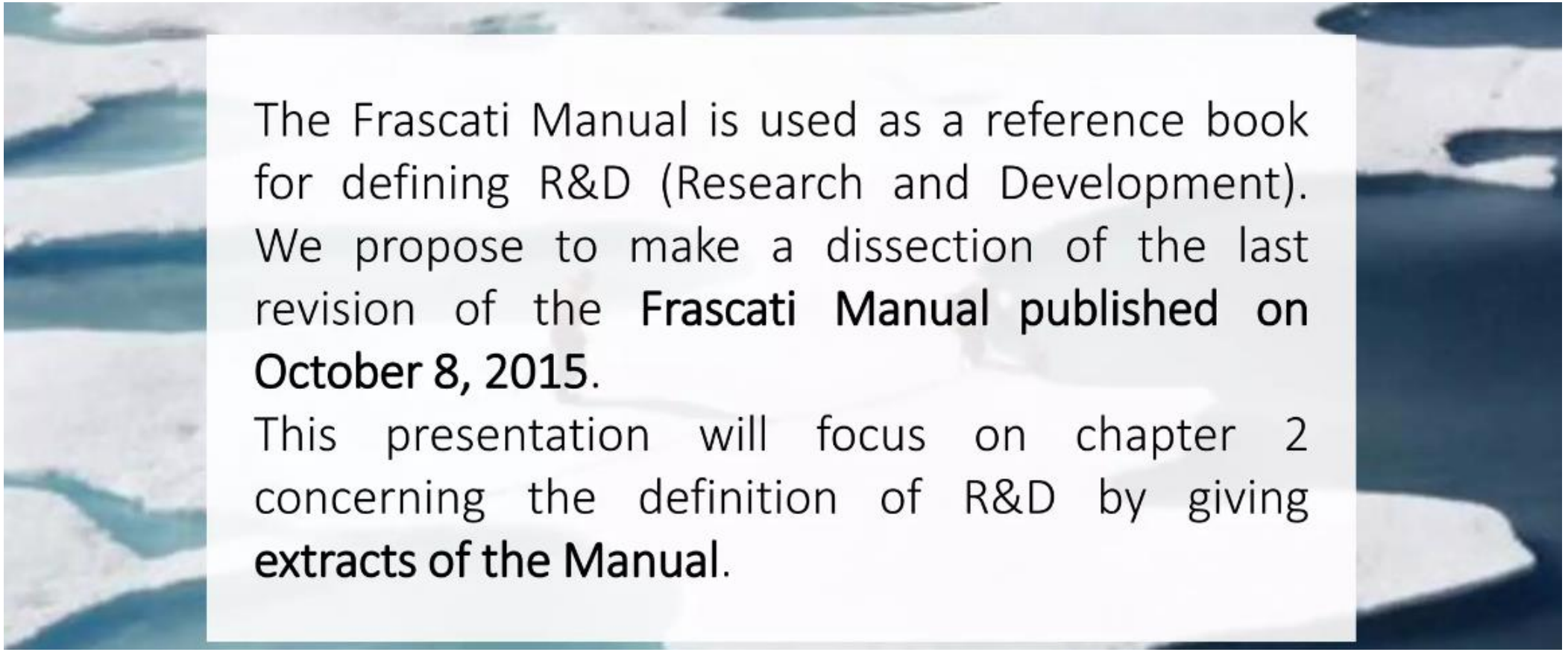
Marmara University, Faculty of Engineering, Department of Bioengineering

25.03.2025



https://www.slideshare.net/Leyton_Group/the-definition-of-rd-following-the-frascati-manual

https://www.slideshare.net/innovationoecd/oslo-manual-2018



The Frascati Manual is used as a reference book for defining R&D (Research and Development). We propose to make a dissection of the last revision of the **Frascati Manual published on October 8, 2015**.

This presentation will focus on chapter 2 concerning the definition of R&D by giving **extracts of the Manual**.


THE TERM R&D COVERS
3 TYPES OF ACTIVITY


1
Basic research

Basic research:

Experimental or theoretical work undertaken primarily to acquire new knowledge of the underlying foundation of phenomena and observable facts, without any particular application or use in view.


2 Applied research

2

Applied research:

Original investigation undertaken in order **to acquire new knowledge**. It is, however, directed primarily **towards a specific, practical aim or objective**.


3
Experimental development



Experimental development:

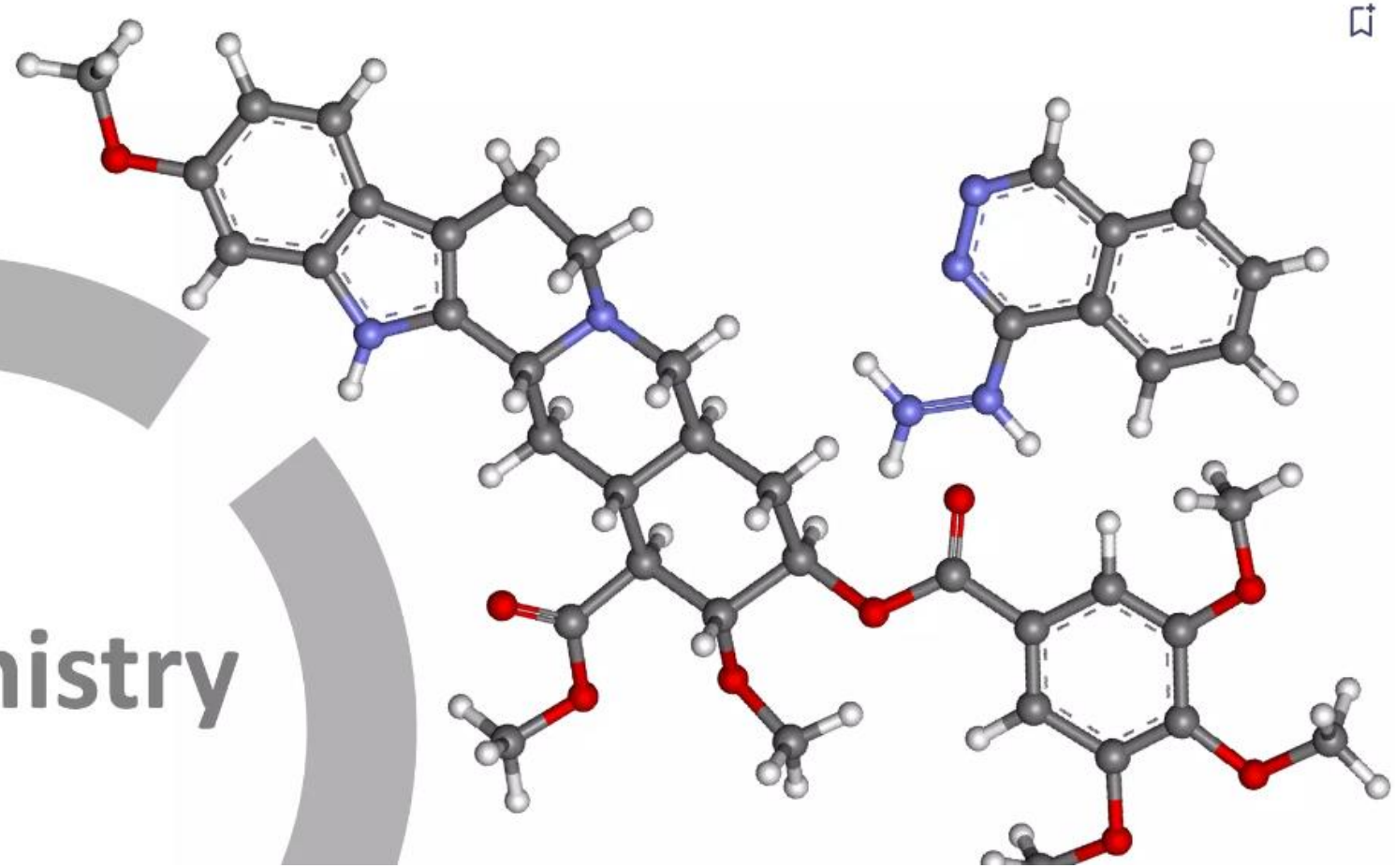
Systematic work, drawing on knowledge gained from research and practical experience and producing additional knowledge, which is directed to producing new products* or processes* or to improving existing products or processes.

*Product refers to a good or a service.

*Process refers to the transformation of inputs to outputs and to their delivery or to organizational structures or practices.

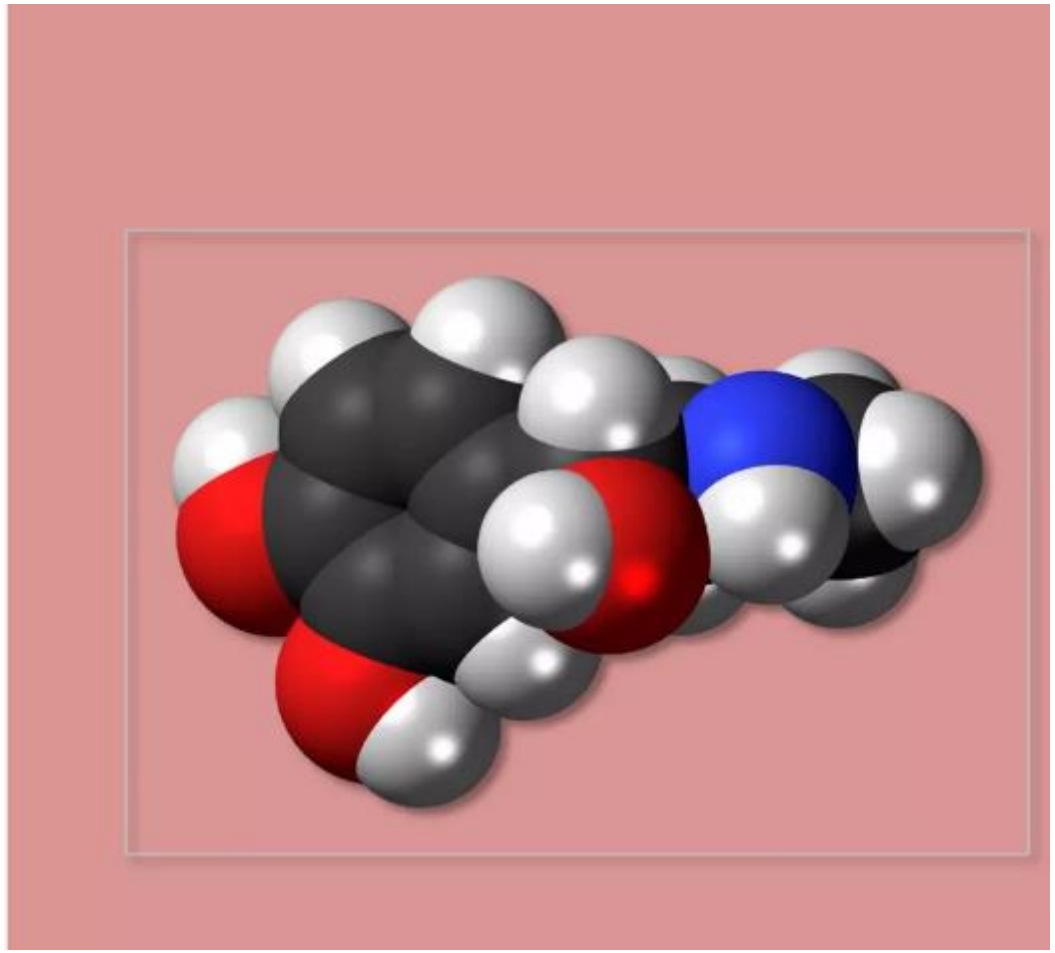

SOME EXAMPLES GIVEN IN
THE FRASCATI MANUAL

Chemistry



Basic Research

Study of a given class of polymerization reactions under various conditions.



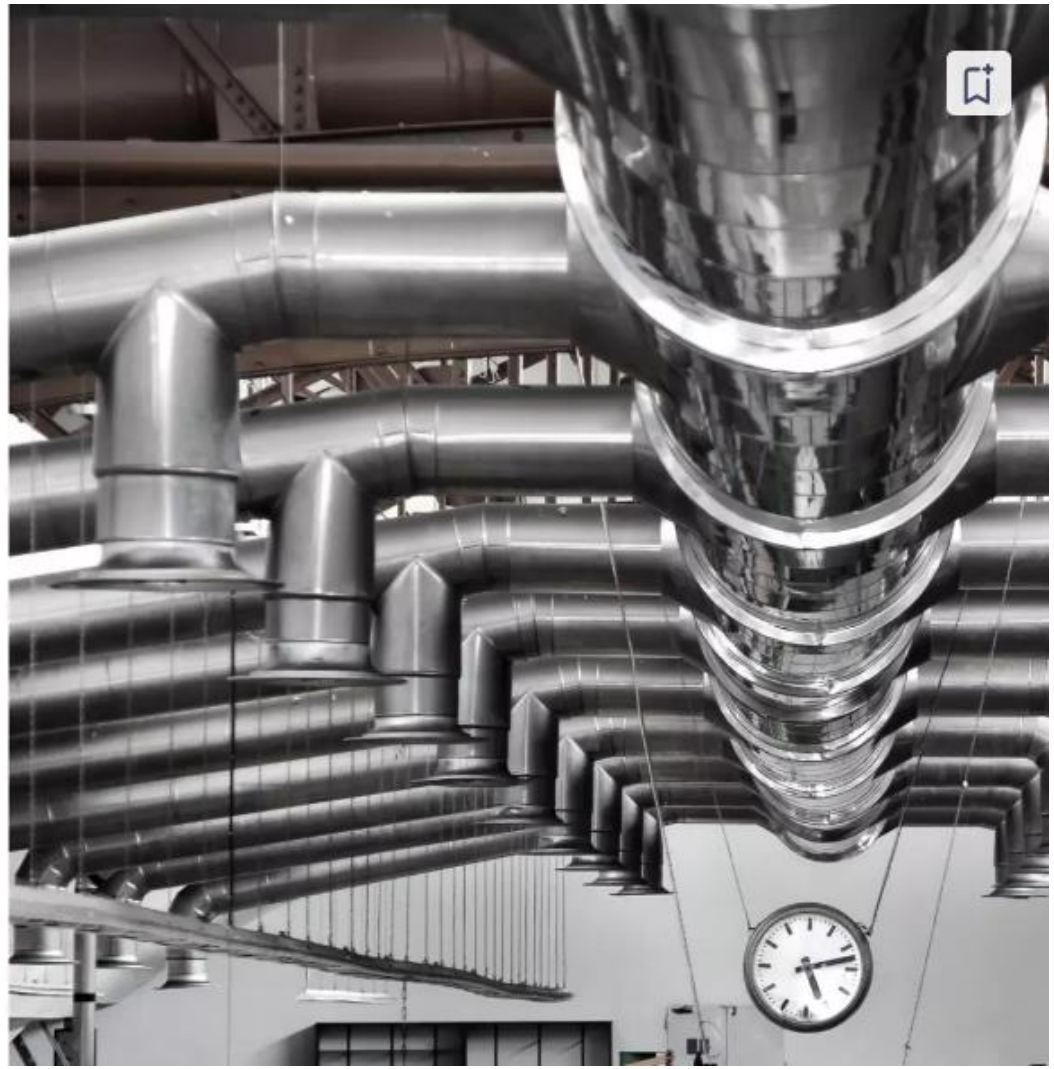


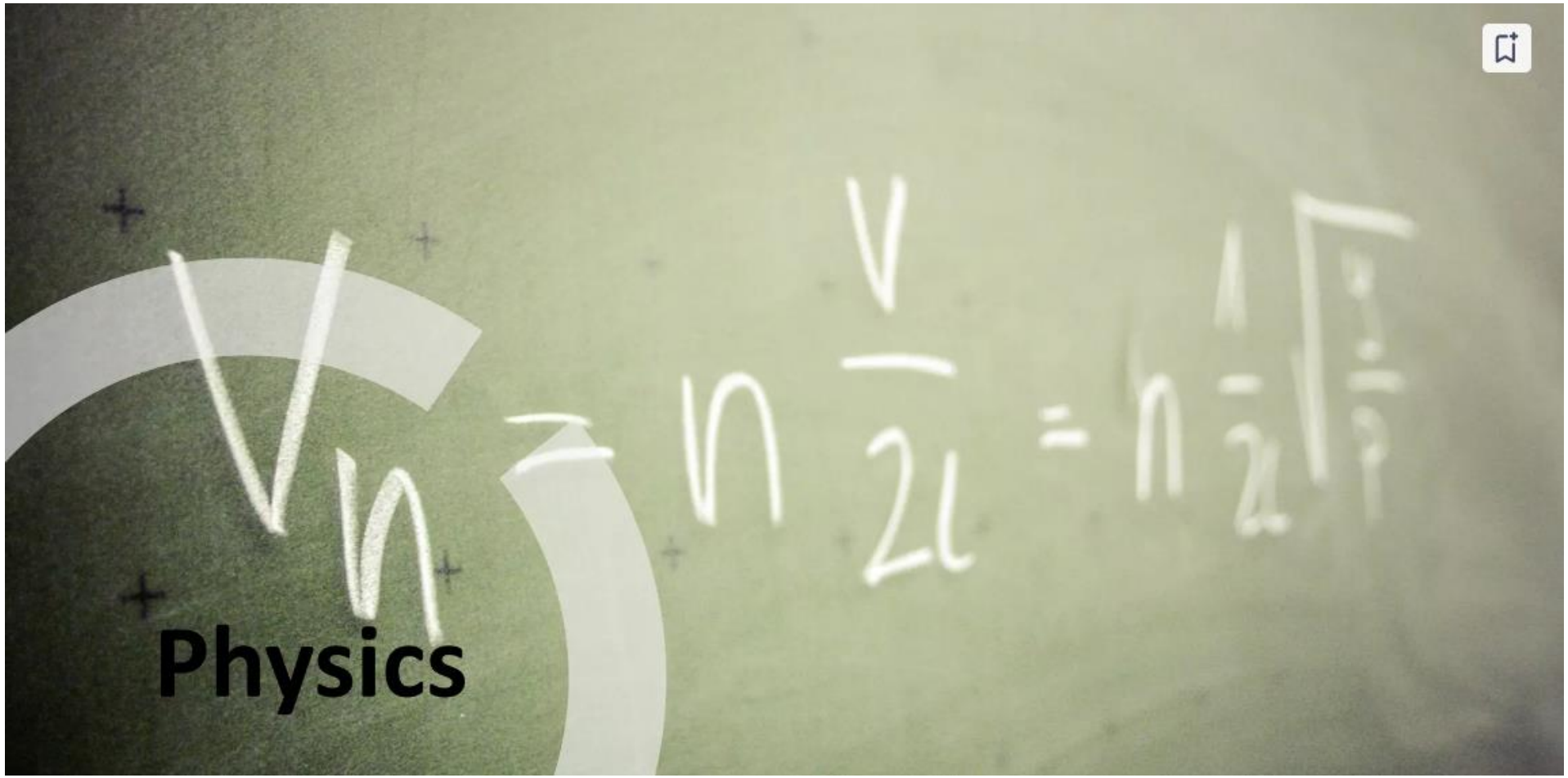
Applied Research

Optimization of a reaction with respect to the production of polymers with given physical or mechanical properties.

Experimental Development

Scaling up of the process that has been optimized at the laboratory level and investigating and evaluating possible methods of producing the polymer as well as products to be made from it.

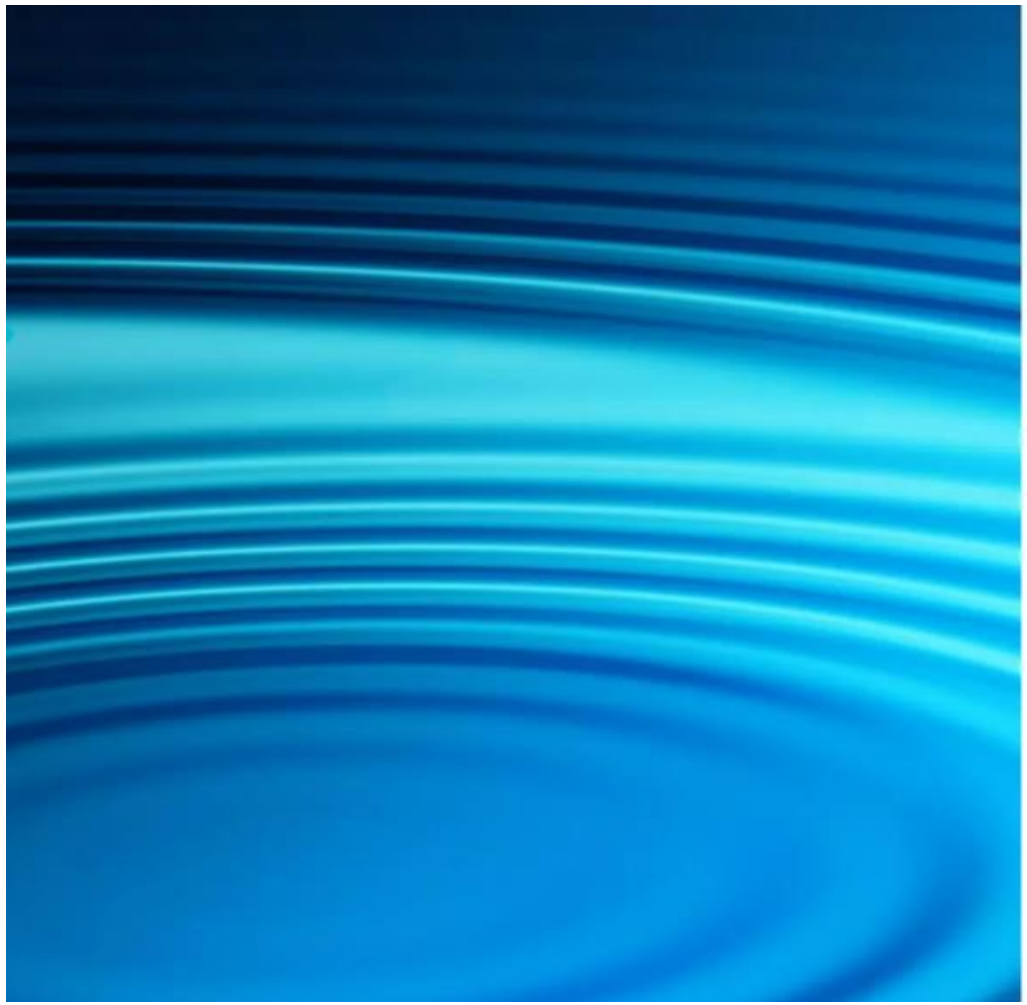



Physics

Basic Research

Modelling of a crystal's absorption of electromagnetic radiation.



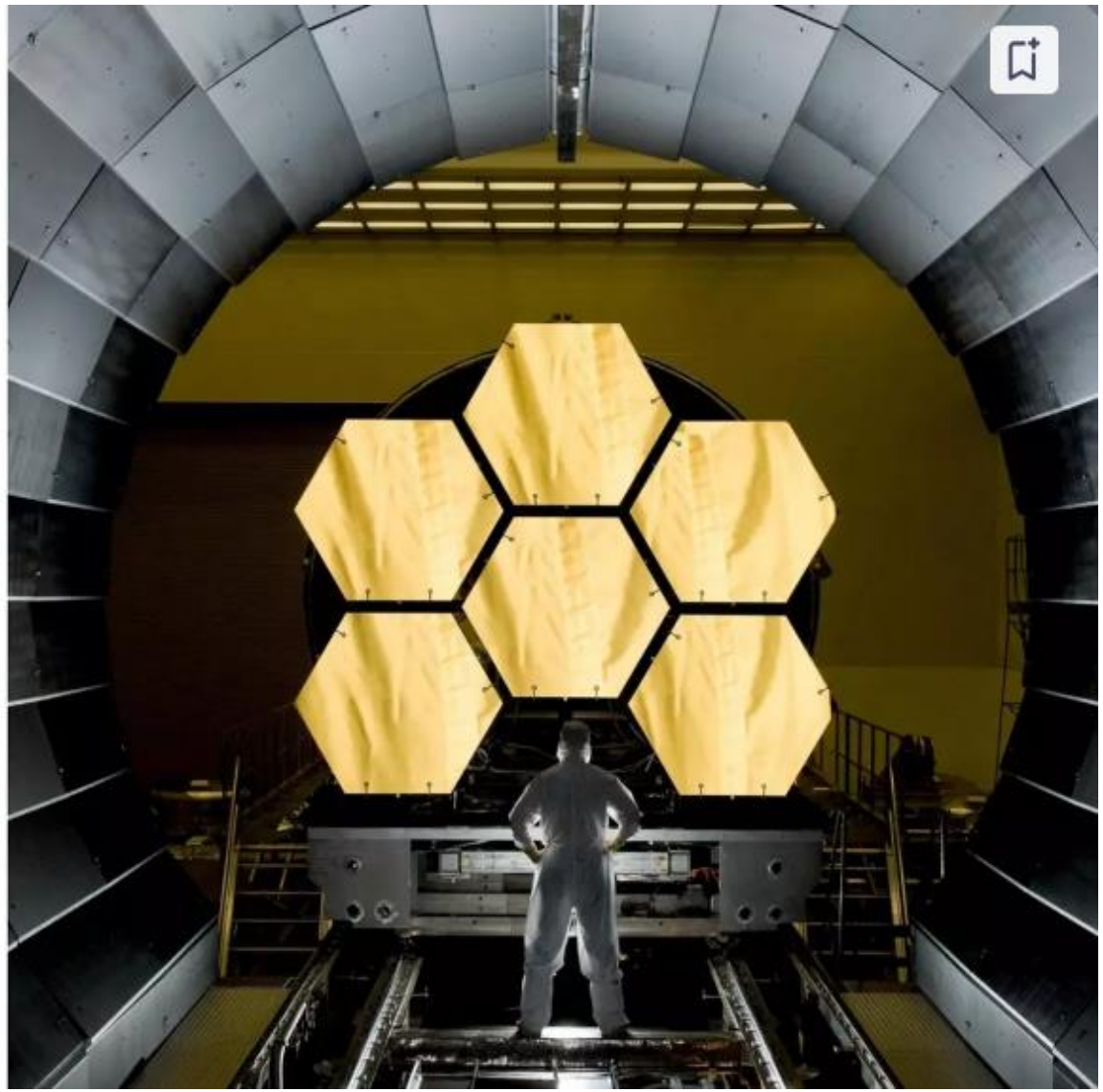


Applied Research

Study of the absorption of electromagnetic radiation by this material under varying conditions of temperatures, impurities, concentration,... to obtain given properties (sensitivity, rapidity, etc.) of radiation detection.

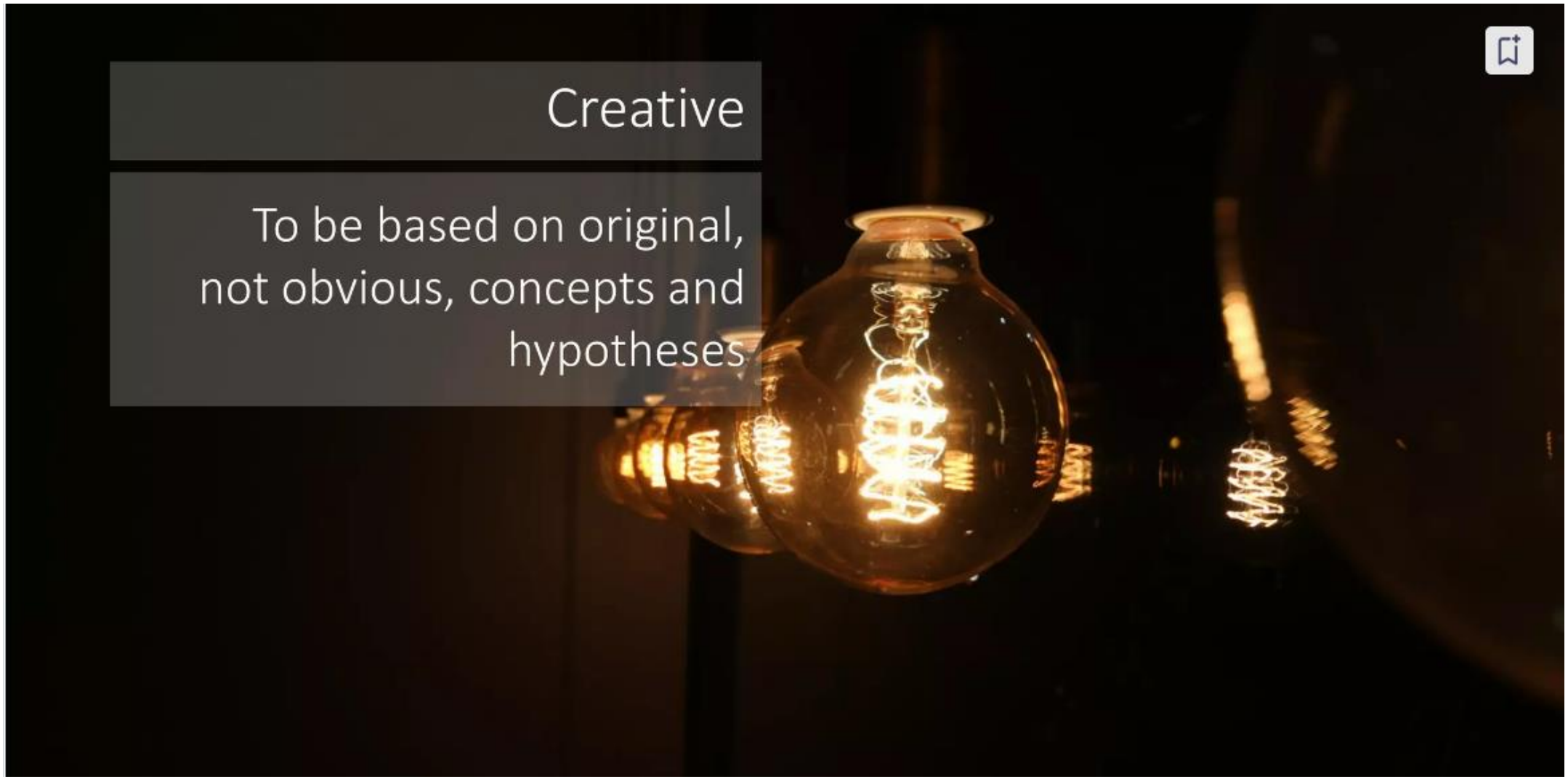
Experimental Development

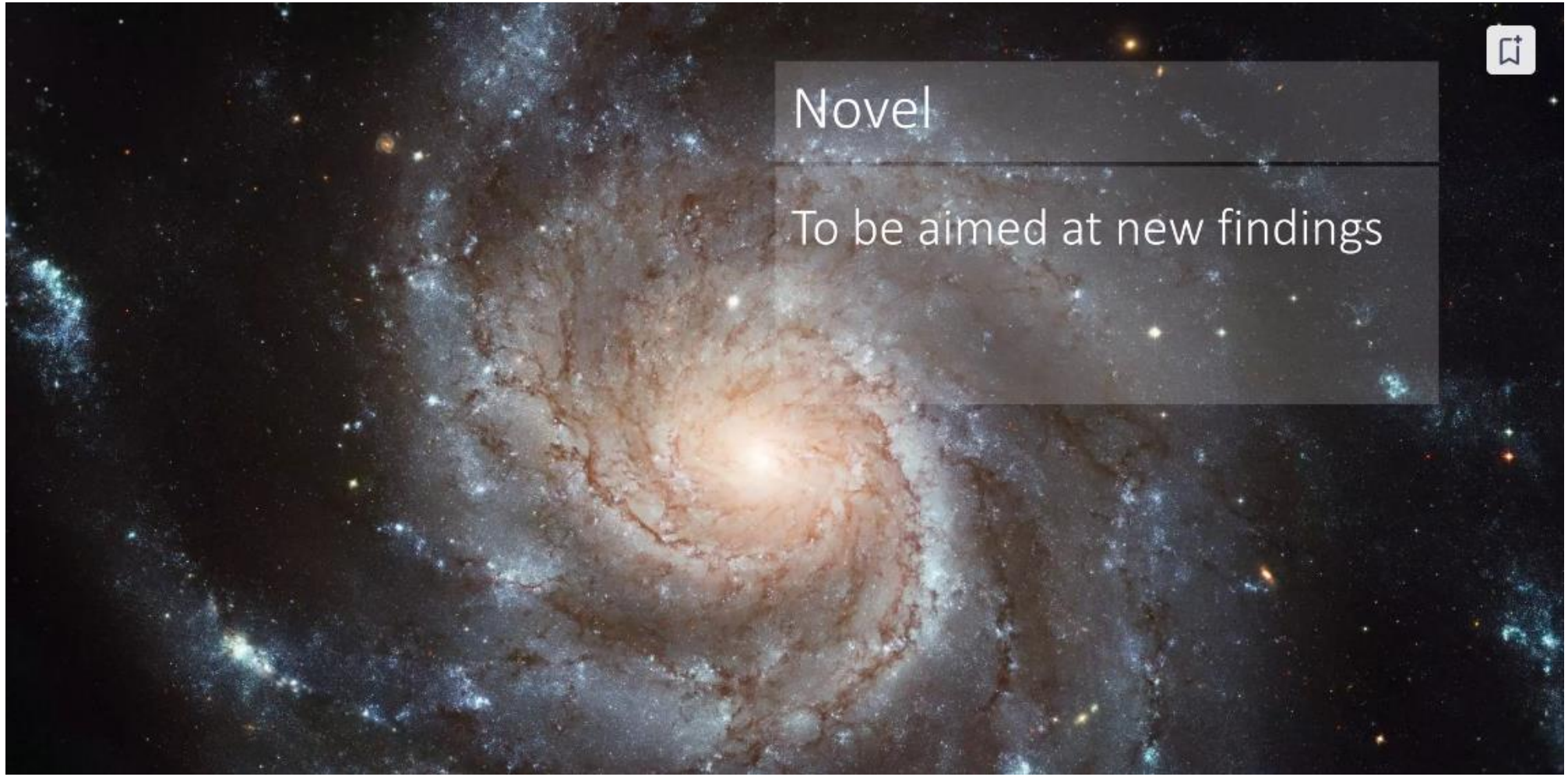
Testing a new device using this material in order to obtain a better detector of radiation than those already existing.



What is and what is not R&D activity by Frascati;




Creative
To be based on original, not obvious, concepts and hypotheses


Novel
To be aimed at new findings


Uncertain about its final outcome (or at least about the quantity of time and resources needed to achieve it)


Transferable and/or reproducible
To lead to results that could be possibly reproduced


Systematic
To be planned and budgeted

Web site: http://oe.cd/frascati

oecd.org/innovation/frascati-manual-2015-9789264239012-en.htm

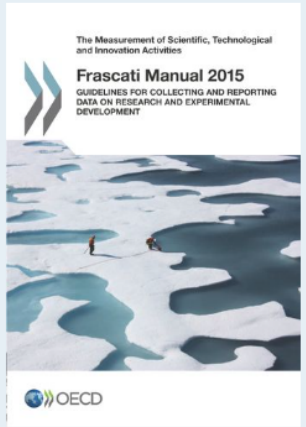
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Frascati Manual 2015

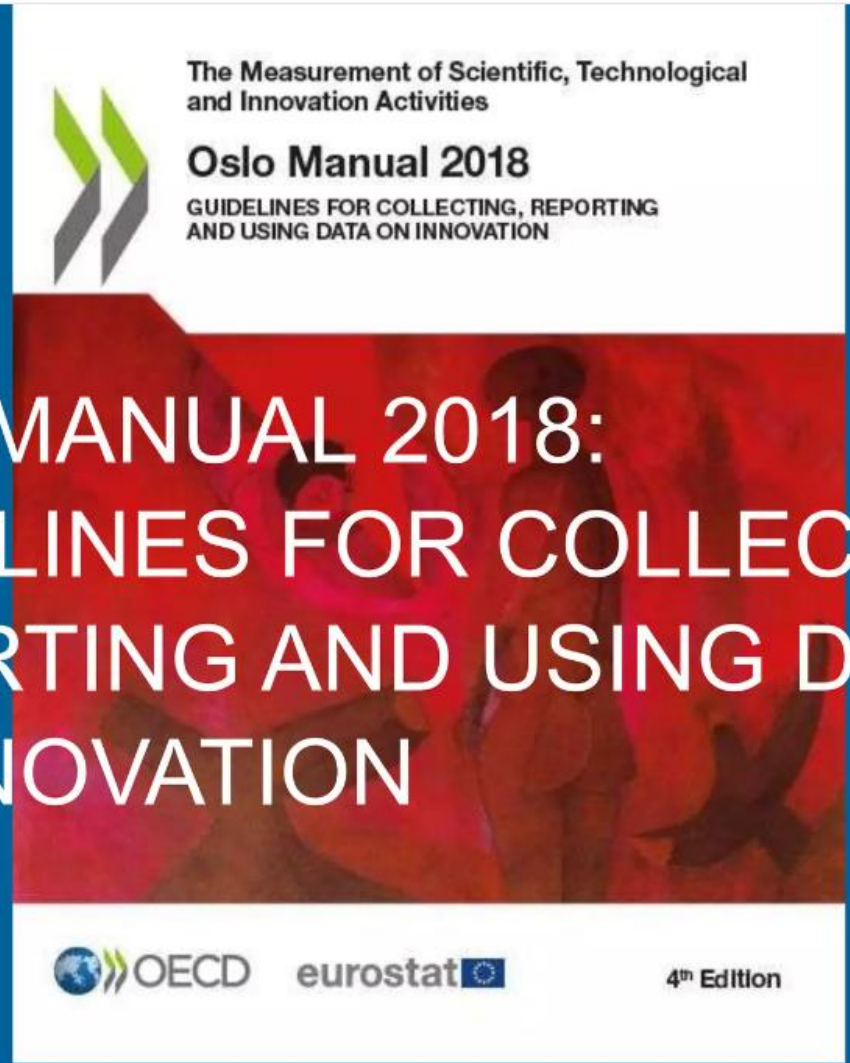
Guidelines for Collecting and Reporting Data on Research and Experimental Development

The internationally recognised methodology for collecting and using R&D statistics, the OECD's Frascati Manual is an essential tool for statisticians and science and innovation policy makers worldwide. It includes definitions of basic concepts, data collection guidelines, and classifications for compiling R&D statistics. This updated edition contains improved guidelines reflecting recent changes in the way R&D takes place and is funded and the wider use of R&D statistics and definitions. It provides new chapters dedicated to the pratical aspects of collecting R&D data in different sectors, as well as new guidan… More

Published on October 08, 2015 Also available in: German, Spanish, French, Korean, Lithuanian, Polish

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OSLO MANUAL 2018: GUIDELINES FOR COLLECTING, REPORTING AND USING DATA ON INNOVATION

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Part I. Introduction to the measurement of innovation

1. Introduction to innovation statistics and the Oslo Manual
2. Concepts for measuring innovation

Part II. Framework and guidelines for measuring business innovation

3. Concepts and definitions for measuring business innovation
4. Measuring business innovation activities
5. Measuring business capabilities for innovation
6. Business innovation and knowledge flows
7. Measuring external factors influencing innovation in firms
8. Objectives and outcomes of business innovation

Part III. Methods for collecting, analysing and reporting statistics on business innovation

9. Methods for collecting data on business innovation
10. The object method for innovation measurement
11. Use of innovation data for statistical indicators and analysis

Glossary of terms

Conceptualising innovation

- Common use of term innovation combines aspects of process (what is done by a subject) and outcome (what comes out)
- Both are relevant and need to be measured.
  - "Outcome" view conceptualised in this manual as: "Innovation(s)"
  - "Process" view conceptualised as "Innovation activity(ies)"

- Innovation
  - More than ideas, implementation required
  - Not necessarily R&D-based
  - At least new to the subject
  - Success is aimed at, but neither guaranteed nor required

General definition of innovation for measurement purposes

"An innovation is

- *a new or improved product or process (or combination thereof)*
- *that differs significantly from the unit's previous products or processes and*
- *that has been made available to potential users (product) or brought into use by the unit (process)."*

The generic term 'unit' describes the actor responsible for innovations. It refers to any institutional unit in any sector, including households and their individual members

Revised innovation definitions for the business sector



*A **business innovation** is a new or improved product or business process (or combination thereof) that differs significantly from the firm's previous products or business processes and that has been introduced on the market or brought into use by the firm.*

***Innovation activities** include all developmental, financial and commercial activities undertaken by a firm that are intended to result in an innovation for the firm.*

Types of business innovation

- *A **product innovation** is a new or improved good or service that differs significantly from the firm's previous goods or services and that has been introduced on the market.*

  -> Goods and services. This edition clarifies role of knowledge products that share features with both.

- *A **business process innovation** is a new or improved business process for one or more business functions that differs significantly from the firm's previous business processes and that has been brought into use by the firm.*

Taxonomy of business processes

- Based on the full array of business functions
- All functions potentially subject to improvements that qualify as innovations

	Business functions devoted to
1.	Production of goods and services
2.	Distribution and logistics
3.	Marketing and sales
4.	Information and communication technology (ICT)
5.	Administration and management
6.	Product and business process development

Innovation activities

Business innovation activities (ch4)

Innovation activities *include all developmental, financial and commercial activities undertaken by a firm that are intended to result in an innovation for the firm.*

A range of knowledge-based activities highlighted for identification as potential loci of innovation activity:

- *Research and experimental development (R&D) activities*
- *Engineering, design and other creative work activities*
- *Marketing and brand equity activities*
- *Intellectual property (IP) related activities*
- *Employee training activities*
- *Software development and database activities*
- *Activities relating to the acquisition or lease of tangible assets*
- *Innovation management activities*

Identify also when activity carried out by another party.

Identify expenditures associated to those items – total (KBC estimates) and separating the amounts incurred in pursuit of innovation.

Understanding internal and external factors shaping innovation in the firm and outcomes

Business innovation capabilities (ch5)

- Management
- IP
- Human resources
- Technology in broad sense (technical, design, user engagement, digital)

External factors influencing business innovation (ch7)

- Markets
  - Position in value chain, supply, demand, platforms
- Public policy – including government support
- Society and environmental aspects

Business innovation objectives and outcomes (ch8)

- Objectives – outcomes
- Impacts on market
- Quantitative measures

OECD Innovation data resources

- The Oslo Manual website
  - http://oe.cd/oslomanual
- OECD Innovation Statistics and Indicators
  - http://www.oecd.org/innovation/inno/inno-stats.htm
- Community of practice on innovation data
  - Password protected, for practitioners: https://community.oecd.org/community/oslo-review

- The Oslo Manual website
  – http://oe.cd/oslomanual

oecd.org/science/oslo-manual-2018-9789264304604-en.htm

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Oslo Manual 2018

Guidelines for Collecting, Reporting and Using Data on Innovation, 4th Edition

What is innovation and how should it be measured? Understanding the scale of innovation activities, the characteristics of innovative firms and the internal and systemic factors that can influence innovation is a prerequisite for the pursuit and analysis of policies aimed at fostering innovation. First published in 1992, the Oslo Manual is the international reference guide for collecting and using data on innovation. In this fourth edition, the manual has been updated to take into account a broader range of innovation-related phenomena as well as the experience gained from recent rounds of innovation surveys in OE… More

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